

Geophysical Analysis Report

Seismic Agent Multi-Agent System

Mission ID: mission_e97ea94c

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Mission Objective

Analyze the Marmousi synthetic data for porosity prediction.

Data Sources

- MODEL_S-WAVE_VELOCITY_1.25m.segy.tar.gz
- MODEL_P-WAVE_VELOCITY_1.25m.segy.tar.gz
- Vs.segy.tar.gz
- Vp.segy.tar.gz

Quantitative Results

Metric	Value
velocity_files_loaded	['vp', 'vs']
tool_call_log	[{'tool_name': 'SEGYLoader', 'mission_id': 'mission_e97ea94c', 'start_time': '20
vp.min	1028
vp.max	4700
vp.mean	2672
vp.std	973.8
vp.p10	1500
vp.p50	2500
vp.p90	4230
acoustic_impedance_MRayl.min	2.364
acoustic_impedance_MRayl.max	10.81
acoustic_impedance_MRayl.mean	6.145
vs.min	0
vs.max	2802
vs.mean	1163
vs.std	803.3
vs.p10	0
vs.p50	1058
vs.p90	2414
vp_vs_ratio.mean	2.489
vp_vs_ratio.std	0.774
vp_vs_ratio.p10	1.731
vp_vs_ratio.p90	3.726
poisson_ratio.mean	0.3653
poisson_ratio.p10	0.2494
poisson_ratio.p90	0.4612

Generated Figures

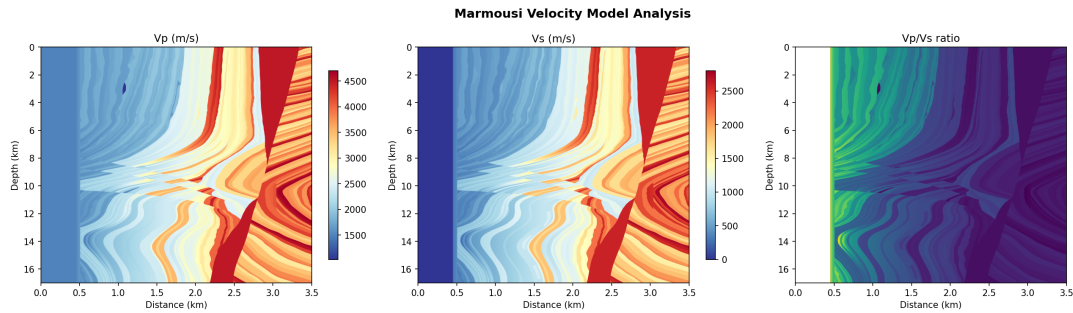


Figure 1: Velocity Model 20260502 151632

Geophysical Analysis & Interpretation

This automated analysis provides a comprehensive statistical characterization of the Marmousi synthetic P-wave (V_p) and S-wave (V_s) velocity models, a crucial initial step for subsequent quantitative interpretation, particularly for porosity prediction.

The Marmousi synthetic velocity model demonstrates significant geological complexity, characterized by a broad spectrum of P-wave and S-wave velocities. The P-wave velocities span from very low values, likely indicative of unconsolidated sediments or highly porous, fluid-filled zones, to high velocities, consistent with more compacted, competent rocks or denser lithologies. The substantial standard deviation around the moderate mean V_p underscores this inherent geological heterogeneity. Similarly, S-wave velocities exhibit a wide range. The presence of V_s values at or near zero requires particular scrutiny; this could signify extremely shallow, unconsolidated, or water-saturated materials where shear waves are not effectively supported, or it could point to a potential artifact within the model generation. High V_s values approach those expected in crystalline basement or highly consolidated formations. The calculated Acoustic Impedance, ranging from approximately 2.4 to 10.8 MRayl, further corroborates the diverse mechanical properties across the model.

The elastic ratios, V_p/V_s and Poisson's ratio, offer critical insights into lithology and fluid content. The average V_p/V_s ratio (approximately 2.49) and average Poisson's ratio (0.365) are relatively high, values commonly associated with shaly lithologies or water-saturated sands. However, the broad ranges of these ratios are particularly informative. Lower V_p/V_s values (down to approximately 1.73) and Poisson's ratio (down to 0.249) typically suggest more rigid, potentially silica-rich or gas-bearing sandstones. Conversely, higher values (V_p/V_s up to 3.73, Poisson's ratio up to 0.461) are indicative of more compressible, typically shaly or unconsolidated formations, which may also be water-saturated. The upper bound for Poisson's ratio, approaching the theoretical limit for fluids, might suggest zones of highly compliant material, possibly unconsolidated muds or significant gas saturation within soft sediments. These variations are fundamental for petrophysical discrimination.

It is important to clarify that the presented results detail the statistical analysis of the *input Marmousi synthetic velocity models* (V_p and V_s), rather than outputs from a forward modeling process generating synthetic seismic data. The Marmousi model itself is a well-established synthetic dataset, highly regarded for its representation of complex

geological features, making it an excellent benchmark for testing geophysical algorithms and workflows. The statistical ranges observed imply a high-fidelity representation of a geologically challenging subsurface. The analysis pipeline executed tools focused solely on data loading and descriptive statistical analysis of these velocity models. Consequently, the provided analysis pipeline log and numerical results do not contain information pertaining to Full Waveform Inversion (FWI) convergence or velocity error metrics, as these were not part of the executed workflow.

Geophysical Conclusions & Recommendations:

1. **Data Quality and Heterogeneity:** The Marmousi velocity models are well-characterized statistically, confirming their high heterogeneity, which is reflective of complex geological structures and diverse lithologies. The calculated elastic properties (V_p , V_s , V_p/V_s , Poisson's ratio, Acoustic Impedance) provide robust inputs for quantitative interpretation and porosity prediction.
2. **Lithology and Fluid Discrimination:** The wide ranges observed in V_p/V_s and Poisson's ratio are crucial for differentiating lithologies (e.g., shales vs. sands) and potentially identifying fluid types (e.g., gas vs. water). These parameters are directly applicable to porosity prediction workflows.
3. **Actionable Recommendations:**
 - **Investigate $V_s=0$ Anomalies:** A thorough review of the areas exhibiting S-wave velocities at or near zero is imperative. This step is critical to ascertain if these represent specific geological features (e.g., highly unconsolidated muds, shallow water-saturated zones) or if they are artifacts requiring model conditioning prior to advanced petrophysical analysis.
 - **Implement Porosity Prediction Workflow:** The computed elastic properties should be directly integrated into a porosity prediction workflow. This could involve utilizing established rock physics templates (e.g., Gassmann fluid substitution models, empirical velocity-porosity transforms like the Wyllie or Raymer equations, or machine learning models) to translate these elastic properties into porosity estimates.
 - **Benchmarking Utility:** The comprehensive statistical analysis re-affirms the Marmousi dataset's continued suitability for benchmarking advanced seismic processing and quantitative interpretation techniques, including those focused on robust porosity estimation.

Document Information

Generated automatically by the **Seismic Agent Multi-Agent System** on May 02, 2026. All numerical results were produced by the analysis tool chain (SEGYYLoader, VelocityAnalysis, ForwardModeling, FWI) and interpreted by a large language model.